DONG BO MARINE ENGINE ROLLER CHAIN

Marine Diesel Engine
Camshaft and Auxiliary Drives
maintenance manual

DONG BO CHAIN IND. CO., LTD.

Dong Bo Marine Engine Roller Chain which meet British Standard(BS) and is uniquely designed to be capable of driving marine diesel engines is designed with most advanced manufacturing technique and the high quality steel.

Dong Bo Marine Engine Roller Chain is used for Camshaft Drives, Fuel Pump Drives and various kinds of auxiliary operated equipment. MAN B&W use it in the marine diesel engine, and Sulzer and Mitsubishi, just to name a few. This catalog shows the items that can be repaired and maintained.

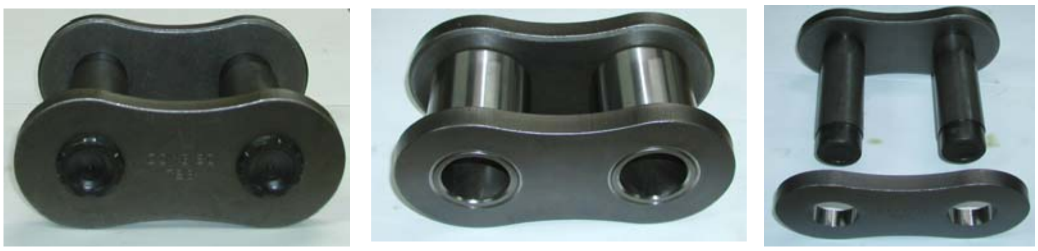
1. Composition

In picture 1. Link plate, pin, bush and rollers are the main compositions of Marine Engine Roller Chain. Link plate supports the tensile stress. Pin and bush builds up the structure of the bearing area. The roller not only absorbs the impact, which is transmitted to the bush, but also prevents the wearing of the sprocket.

Picture 1

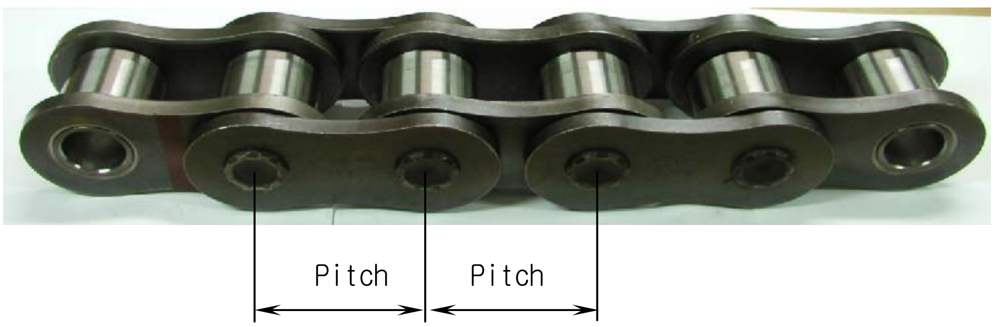
Chain is composed of two pins and two pin link plates as in picture 2 and pin is assembled into link plate by press-fit. In roller link two bushes are pushed into link plate, and roller is assembled into the outer bush to make roller turn freely. The connecting link is supplied a press-fit outer plate.

Pin link Roller link Connecting link



Picture 2

Look at the picture 3. Chain is connected and assembled into roller link and pin link plate secured by riveting. The length of chain is measured by roller centers to roller centers, the chain in picture 3 is five pitches long, commence inner link finish inner link.

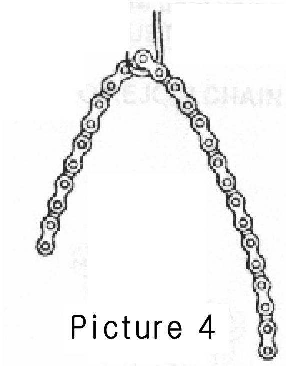


Picture 3

2. Instructions & Installation

2-1. Unpacking

The chain could be heavy, so put packing case on its side, then open and carefully lift chain out vertically by lifting device. Picture 4.

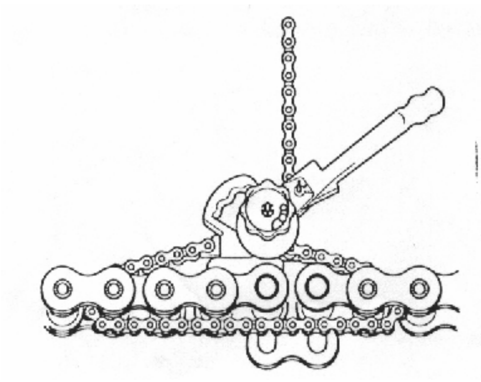


Picture 4

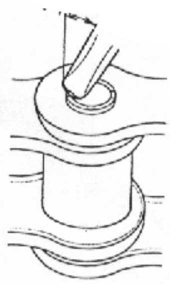
2-2. Installation

If necessary use a lifting device to put the chain on to the engine. Picture 4. It may require a pulling device, picture 5, to pull the roller links together. Put the connecting links pins into the roller link bushings and then push on connecting link top plate, push a little on each pin at a time so that the plate does not bend. The connecting link pins must then be riveted with a round punch. Picture 6.

NOTE: Do not change the connecting link plate holes or pin diameter to assembly connecting link plate and pin easily.



Picture 5

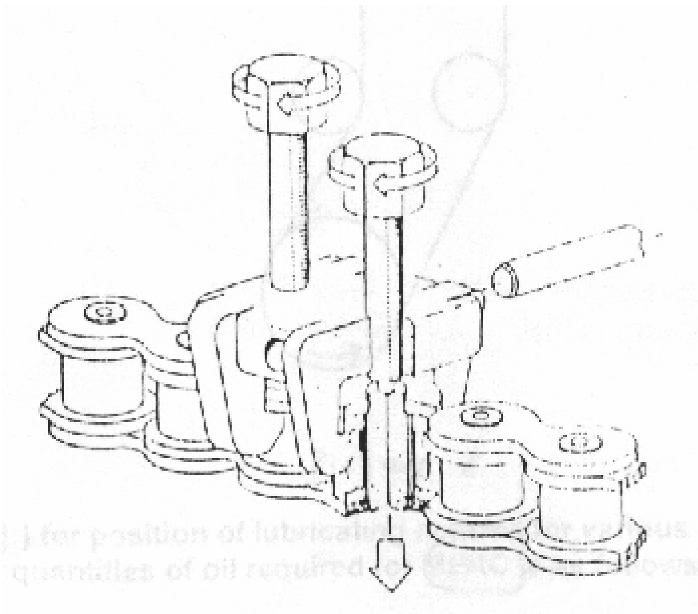


Picture 6

2-3. Disassembly

If the chain has to be disassembled, the connecting link has to be found. It is usually marked red in color. Using tool, Picture 7, tighten up the bolt on to center of the pin and tighten, pushing the pin through outer plate and into the bush. Move and put on the second pin and repeat procedure until you can pull out connecting link pins. **REMEMBER** this connecting link **MUST NOT** be used again as all interference fits have been destroyed.

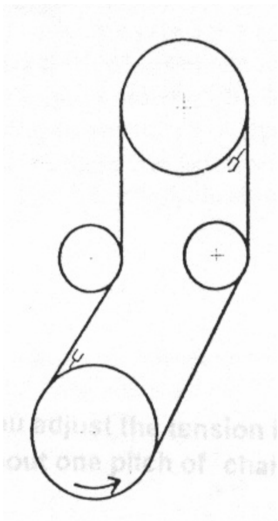
A NEW LINK MUST BE USED TO REJOIN CHAIN.



Picture 7

3. Lubrication

Lubrication is very important for High Quality Roller Chains. It helps prevent wear stretching and creates a film of oil between pin, bush and roller. It also helps reduce noise at high speed. Locate the lubrication as close to the chain entering the sprockets as possible. Picture 8.



Picture 8

See picture 9 for position of lubricating nozzles for various types of chain.

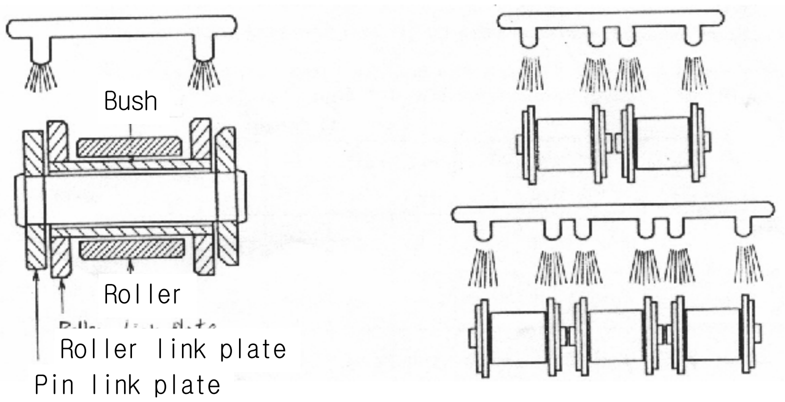
The minimum quantities of oil required for Marine Engine Roller Chain is as follows:

Single strand chain: 10 Liters(2 gals.) per inch length of chain/ hour

Double strand chain: 20 Liters(4 gals.) per inch length of chain/hour

Triple strand chain: 30 Liters(6 gals.) per inch length of chain/hour

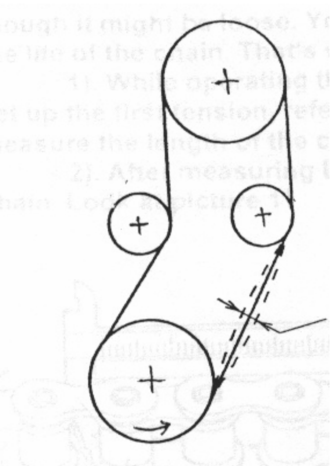
NOTE: Check nozzles and pipes regularly blockages.



Picture 9

4. Chain tension and Adjustment marine engine

Vibrations are usually caused by the chains becoming loose by stretching and wear. These vibrations can be reduced as you change the tension. Engine manufactures specifications should be checked. You should make the first tension a half of the tension, which is affected in the chain while driving. If it is difficult to measure the tension you should adjust it. See picture 10.



You adjust the tension in the center to make the movement about one pitch of the chain by pushing and pulling it.

Picture 10

After shop test and sea trial test the chain which is used first may be expanded for about 0.02~0.05%. So you should adjust the chains tension and must not loosen it. As the tension is adjusted, you won't be able to see the loose chain, and you must tighten up the loose chain every 4000 hours.

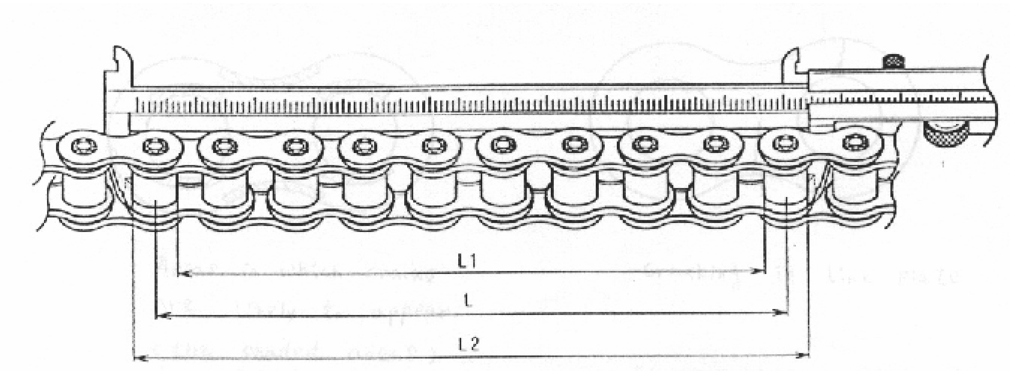
5. Inspection & Repairs

Whenever the link is bent owing to the contact of the chain and the sprocket, the chain might loosen with the wear of the pin and bush. Because Dong Bo Marine Engine Roller Chain is a much higher quality then standard manufactured roller chain the wear will be less and also run better.

The inspection and repairs is exactly needed for the complete life of the chain. As the chain loosens, you can see the wear of the pin and the surface of bush. The chain for a marine diesel engine must not be exchanged for 10 years, even though it might be loose. You must measure the loose length regularly to check the life of the chain. That's why both of them are in inverse proportion.

(1) While operating the engine slowly, you tighten up the loose chain and set up the first tension, referring to section 4. After stopping the engine, you measure the length of the chain where the tension is set up.

(2) After measuring L1 and L2, you calculate the real length L of the chain. Look at picture 11.



Picture 11

(3) You measure it more than 6 pitches to minimize the measurement error.

(4) In the Table 1 the loose maximum tolerance is 1% longer than the standard length. If the measured length exceeds the maximal allowable length you must change the chain.

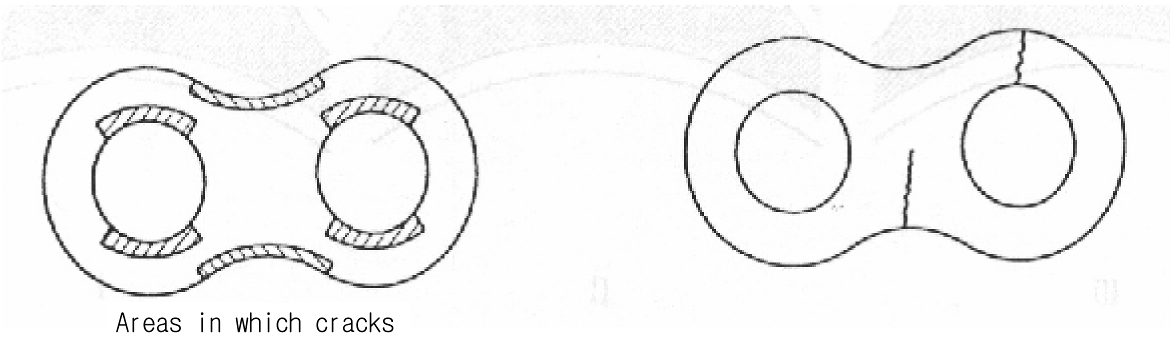
Unit : mm

Chain No.	16B	20B	24B	28B	32B	40B	48B	56B	64B	72B
Pitch(inch)	1.0	1.25	1.5	1.75	2.0	2.5	3.0	3.5	4.0	4.5
Standard length (Pitch x 6 Links)	152.4	190.5	228.6	266.7	304.8	381.0	457.2	533.4	609.6	685.8
Max. allowable length (For 6 Links)	153.9	192.4	230.9	269.4	307.9	384.8	461.8	538.7	615.7	692.7
Standard length (Pitch x 10 Links)	254.0	317.5	381.0	444.5	508.0	635.0	762.0	889.0	1016.0	1143.0
Max. allowable length (For 10 Links)	256.5	320.7	384.8	449.0	513.1	641.4	769.6	897.9	1026.2	1154.4

Table 1.

Every parts from Dong Bo Marine Engine Roller Chain is produced through the ultrasonic flaw detecting test(UT TEST) and the magnetic particle test (MT TEST). However, if the vibration of the chain, witch comes from the drive exceeds the fatigue(endurance) limit, the chain can be broken. In case the chain vibrates extremely, cracks can happen on the link plate.

Therefore check the chain very carefully.
The crack on the link plate shows the first sign that the fatigue limit exceeded the chain. Generally, the crack is a slow progress. So examine the link plate carefully and you will prevent an accident.
If you do find a crack on the link plate, even if it is a small one, the whole chain **MUST** be changed. That is because every link gets weakened on the same condition. Picture 12.



Areas in which cracks are likely to appear (the shaded areas)

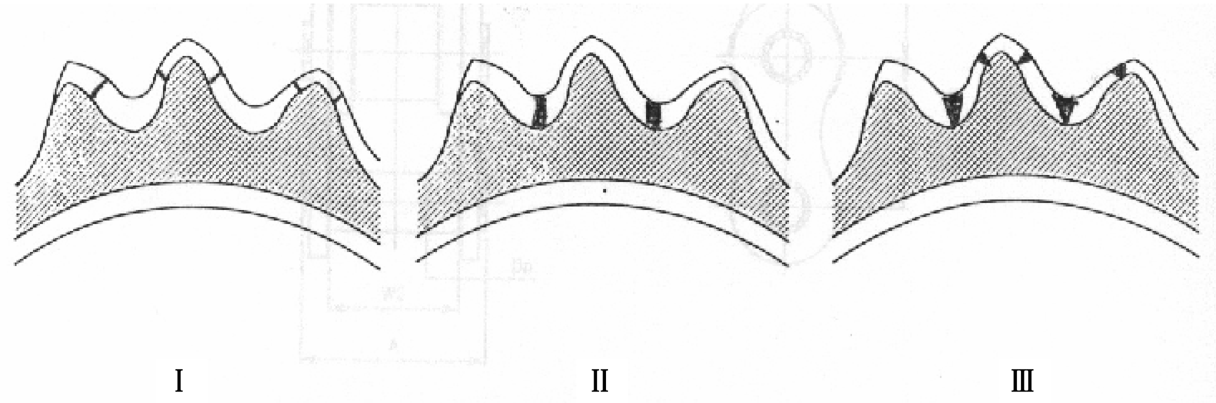
Cracking in link plate

Picture 12

The roller doesn't cause the problem by the impact; it is the severe vibration that makes the crack happen on the rollers. You must check for cracks on the rollers often.
To prevent the fatigue fracture you must minimize the vibration of the chain as much as possible. If you adjust the tension of the chain and can not stop the vibrating, you must discuss with the engine manufacturer.

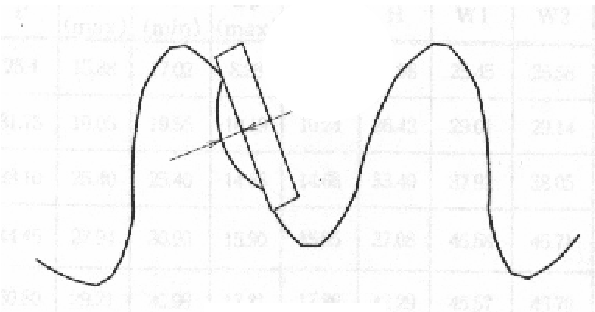
6.The Sprocket Inspection

To examine whether the chain roller and sprocket teeth exactly line up, you should check the contact trace and the part of the wear in the inner teeth after SHOP TEST.
If they precisely line up, the width of the contact surface will be fixed as Ⅰ or Ⅱ in the Picture 13. If the contact surface isn't level the roller and sprockets are nit in a straight line as shown in Ⅲ. Therefore you must put them in line.
On the whole the roller doesn't make contact between the sprocket teeth. But if the chain roller is tensioned on the loose part, you can see the were from this.
When idler or tightener is tilted, the contact point will be the bottom between the teeth as Ⅱ in the Picture 13.



Picture 13

The sprocket teeth also can be worn while contacting the chain roller continually. Picture 14
There is a maximal wear tolerance of the teeth in the Table 2.



Picture 14

Chain No.	16B	20B	24B	28B	32B	40B	48B	56B	64B	72B
Pitch(inch)	1.0	1.25	1.5	1.75	2.0	2.5	3.0	3.5	4.0	4.5
Maximum allowable wear(mm)	0.9	1.1	1.4	1.5	1.6	2.3	2.5	3.0	3.6	4.1

Table 2.

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